A Comparative Study of Image Dehazing Algorithms

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by

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<u>CERTIFICATE</u>

*This is to certify that the work contained in this thesis entitled "**A Comparative Study of Image Dehazing Algorithms**" is a bonafide work of **Adarsh Kumar** (**Roll No. 220122004**), carried out in the Department of Mehta Family School of Data Science and Artificial Intelligence, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

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CERTIFICATE

I hereby declare that the work presented in this project titled **'A Comparative Study of Image Dehazing Algorithms"** submitted to **Assistant Professor Dr. Teena Sharma**, **Indian Institute of Technology Guwahati**, for the project report of BTP, is my original work. I have not plagiarized or submitted the same work for any other project report, nor have I used any LLM or AI tools in the preparation of this document.

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1.1 Abstract

Single image dehazing is a critical challenge in computer vision system. This review comprehensively discusses the different techniques for image dehazing. These techniques are broadly classified into 4 categories according to their techniques which are Physics based, CNN based, Encoder Decoder based and Transformer based. Physics-based methods, represented by the Dark Channel Prior and Color Attenuation Prior, make use of atmospheric scattering models like Narmsimhan and Nayar [10, 11] Equation. CNN based models like DehazeNet [3] and DCNet [5] use CNN to dehaze image. Encoder-decoder architectures further improve feature extraction and reconstruction through skip connections and multi-scale learning. Recent Transformer based frameworks like Dehazeformer [8] employs self attention to dehaze image. This work summarizes the architectural evolution from physically inspired formulations toward data-driven and context-aware networks, underlining the improvements in performance and generalization ability.

1.2 Introduction

Outdoor images taken in bad weather conditions, like haze, fog, or smoke, typically present low visibility, low contrast, and color distortion. It occurs by reason of the light scattering and absorption effects related to atmospheric particles; this is what causes attenuation of scene radiance before it reaches the imaging sensor. This constitutes a key challenge for computer vision systems, such as autonomous driving, remote sensing, and surveillance, where reliable visual perception is quite important. Single image dehazing, namely, the reconstruction of a clear image from one given hazy input, has then become an important research topic within low-level vision.

The image formation model for hazy environments was first formalized by Narasimhan and Nayar [10, 11] (2002), commonly known as the atmospheric scattering model. It describes the observed hazy image $I(x)$ as a combination of the attenuated scene radiance and the airlight component.

$$I(x) = J(x)\, t(x) + A\, (1 - t(x)) \tag{1}$$

where $I(x)$ represents the observed intensity at pixel x, $J(x)$ is the true scene radiance, and A denotes the global atmospheric light.

The transmission map $t(x)$ is modeled as:

$$t(x) = e^{-\beta d(x)} \tag{2}$$

where β is the scattering coefficient and $d(x)$ is the scene depth.

The goal of image dehazing is to estimate $J(x)$ from the observed $I(x)$. This is an *ill-posed inverse problem*, since both A and $t(x)$ are unknown.

To handle this challenge, various methods have been suggested based on physical priors, statistical assumptions, and deep learning architecture. The evolution of single-image dehazing approaches is categorized broadly into four major classes: (i) physics-based methods, (ii) CNN-based models, (iii) encoder–decoder architectures, and (iv) Transformer-based frameworks. Early physics-based methods, including the Dark Channel Prior (DCP) and Color Attenuation Prior (CAP), estimated transmission and atmospheric light based on handcrafted priors and the atmospheric scattering model. While interpretable and computationally efficient, such methods usually cannot handle scenes that contain bright regions or dense haze. Subsequently, CNN-based models, such as DehazeNet [3], AOD-Net [4], and DC-Net [5], introduced the data-driven paradigm by learning haze-relevant features directly from paired hazy and clear images, further improving restoration quality and adaptability. Further developments led to encoder-decoder architectures, which use multi-scale feature extraction and skip connections to preserve structural and texture details while reconstructing the image. These models balance global context understanding with local detail preservation. Recently, Transformer-based approaches have, in fact, redefined the dehazing landscape; TransWeather [7], Dehaze-Former [8],are examples of this, using self-attention mechanisms to capture long-range dependencies and global haze distributions, achieving state-of-the-art performance on challenging benchmarks. Although there are remarkable achievements, several limitations still exist in current approaches regarding domain generalization, real-time inference, and physical interpretability. This paper reviews single-image dehazing techniques in four categories, elaborating on their principles, architectures, and characteristics. It further discusses the shift of paradigm from physics-driven to data-driven and sheds light on promising directions for developing efficient, generalizable, and physically grounded dehazing models.

1.3 Contributions

The major goal of vision-based algorithms is to perceive and interpret visual information from the environment much like human perception. The human visual system intuitively distinguishes between clear and hazy regions and adaptively identifies the different haze densities. Inspired by such a mechanism, image dehazing algorithms have evolved through generations-from physics-based formulations to deep learning and Transformer-based architectures-each outperforming its predecessor in feature representation, learning capability, and generalization.

This paper compares four representative approaches to single-image dehazing, each belonging to a different methodological paradigm. The specific contributions of this work are summarized below.

1.3.1 Physics-Based Method: Dark Channel Prior (DCP)

The dark channel prior by He et al. [1] (2009) is one of the most influential physics-based methods for dehazing. That method operates on the atmospheric scattering model and assumes that in most non-sky patches of haze-free outdoor images, at least one color channel has pixels with very low intensity. Using this prior, the algorithm estimates both the transmission map and global atmospheric light to recover clear scene radiance.

It is interpretable and computationally efficient; however, it tends to overestimate haze in bright or sky-dominant regions and performs poorly under dense haze conditions due to its dependency on local intensity statistics.

1.3.2 CNN-Based Method: Dark Channel Network (DCNet)

This paper proposes a CNN-based architecture, namely the Dark Channel Network (DC-Net [5]), to estimate the true scene transmission from a hazy image and overcome the limitations of handcrafted priors.

The proposed architecture consists of two major components:

Feature Extraction Layer: This layer extracts the haze-relevant spatial features from the input image.

Convolutional Neural Network Layer: Applies trainable convolutional filters in order to predict the transmission map.

The estimated transmission is then integrated within the Narasimhan and Nayar atmospheric scattering model to reconstruct the dehazed image. The proposed DCNet [5] achieves comparable or superior performance with fewer layers than conventional CNN-based dehazing networks like DehazeNet [3] and AOD-Net [4], hence offering a strong trade-off between accuracy and computational efficiency.

1.3.3 Encoder–Decoder Method: GridDehazeNet

As an example of the encoder–decoder-based approach, GridDehazeNet [6] (Liu et al., CVPR 2019) utilizes a U-Net-like architecture for enhancing both global and local haze removal. The method introduces grid connections between encoder and decoder paths that can enable multi-scale feature aggregation and hence better spatial consistency in reconstructed images.

Lightweight, interpretable, and efficient, GridDehazeNet [6] presents a good representative from this architectural family. It effectively recovers the fine textures of the image while keeping the balance of global illumination.

1.3.4 Transformer-Based Method: DehazeFormer

In this work is DehazeFormer [8]. DehazeFormer [8] improves the Swin Transformer backbone by adopting a hierarchical encoder–decoder architecture with deformable self-attention to model global features efficiently. It captures long-range dependencies and performs a cross-scale feature fusion, enabling superior dehazing quality with fine detail reconstruction. Such a model realizes state-of-the-art performance on benchmark datasets like RESIDE and Haze4K, representing the current frontier in data-driven and context-aware dehazing systems.

Methods	ITS		OTS		RESIDE-6K		RS-Haze		Overhead		
	SOTS-indoor		SOTS-outdoor		SOTS-mix		RS-Haze-mix				
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	#Param	MACs	Latency
(TPAMI'10) DCP [4]	16.62	0.818	19.13	0.815	17.88	0.816	17.86	0.734	-	-	-
(TIP'16) DehazeNet [8]	19.82	0.821	24.75	0.927	21.02	0.870	23.16	0.816	0.009M	0.581G	0.919ms
(ECCV'16) MSCNN [9]	19.84	0.833	22.06	0.908	20.31	0.863	22.80	0.823	0.008M	0.525G	0.619ms
(ICCV'17) AOD-Net [11]	20.51	0.816	24.14	0.920	20.27	0.855	24.90	0.830	0.002M	0.115G	0.390ms
(CVPR'18) GFN [12]	22.30	0.880	21.55	0.844	23.52	0.905	29.24	0.910	0.499M	14.94G	3.849ms
(WACV'19) GCANet [14]	30.23	0.980	-	-	25.09	0.923	34.41	0.949	0.702M	18.41G	3.695ms
(ICCV'19) GridDehazeNet [13]	32.16	0.984	30.86	0.982	25.86	0.944	36.40	0.960	0.956M	21.49G	9.905ms
(CVPR'20) MSBDN [17]	33.67	0.985	33.48	0.982	28.56	0.966	38.57	0.965	31.35M	41.54G	13.25ms
(ECCV'20) PFDN [15]	32.68	0.976	-	-	28.15	0.962	36.04	0.955	11.27M	50.46G	4.809ms
(AAAI'20) FFA-Net [18]	36.39	0.989	33.57	0.984	29.96	0.973	39.39	0.969	4.456M	287.8G	55.91ms
(CVPR'21) AECR-Net [19]	37.17	0.990	-	-	28.52	0.964	35.69	0.959	2.611M	52.20G	6.095ms
(ours) DehazeFormer-T	35.15	0.989	**33.71**	0.982	**30.36**	0.973	39.11	0.968	0.686M	6.658G	9.380ms
(ours) DehazeFormer-S	36.82	**0.992**	**34.36**	0.983	**30.62**	**0.976**	**39.57**	**0.970**	1.283M	13.13G	13.44ms
(ours) DehazeFormer-B	**37.84**	**0.994**	**34.95**	0.984	**31.45**	**0.980**	**39.87**	**0.971**	2.514M	25.79G	27.16ms
(ours) DehazeFormer-M	**38.46**	**0.994**	**34.29**	0.983	**30.89**	**0.977**	**39.71**	**0.971**	4.634M	48.64G	25.04ms
(ours) DehazeFormer-L	**40.05**	**0.996**	-	-	-	-	-	-	25.44M	279.7G	54.32ms

Fig. 1 Overview of different dehazing architectures discussed in this paper: (a) Physics-based DCP, (b) CNN-based DCNet, (c) Encoder–Decoder GridDehazeNet, and (d) Transformer-based ViT model.

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1.4 Image Dehazer Transformer: Proposed Architecture

The proposed **Vision Transformer (ViT)** [9] architecture performs end-to-end single-image dehazing using a physics-inspired multi-scale haze representation as input. The network directly restores the haze-free image without explicitly estimating the transmission map or atmospheric light. The overall architecture comprises two major components:

1. **Feature Embedding Layer**
2. **Vision Transformer Encoder–Decoder Layer**

In the following subsections, these layers are discussed in detail.

1.4.1 Feature Embedding Layer

The feature embedding layer transforms the seven-channel haze-aware representation of the input image into patch tokens suitable for Transformer processing. This representation is derived from the *multi-scale dark channel prior (DCP)* and the *Value (V)* channel from the HSV color space.

Multi-Scale Feature Construction

Given an input hazy image $S(a,b) \in \mathbb{R}^{H\times W\times 3}$, the Value (V) channel is extracted as:

$$S_V(a,b) = \max_{\lambda\in\{R,G,B\}} S_\lambda(a,b) \tag{3}$$

This channel captures illumination and edge information crucial for haze estimation. The multi-scale dark channel maps are then computed using local minima over neighborhoods of increasing patch size:

$$D_p(a,b) = \min_{(i,j)\in\Omega_p(a,b)} \left(\min_{\lambda\in\{R,G,B\}} S_\lambda(i,j) \right) \tag{4}$$

where $\Omega_p(a,b)$ denotes a local patch centered at pixel (a,b) with window size $\sqrt{p} \times \sqrt{p}$. For the proposed architecture, dark channel maps are generated for patch sizes $p \in \{1 \times 1, 3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9, 10 \times 10\}$.

The composite multi-scale representation is then obtained by concatenating the V channel and the six dark channel maps:

$$F(a,b,z) \leftarrow [S_V(a,b), D_{1\times1}(a,b), D_{3\times3}(a,b), D_{5\times5}(a,b), D_{7\times7}(a,b), D_{9\times9}(a,b), D_{10\times10}(a,b)] \tag{5}$$

where $z = 7$ denotes the total number of feature maps. This multi-channel representation encodes haze intensity and illumination cues at multiple spatial scales, serving as an enriched input for the Vision Transformer.

1.4.2 Vision Transformer Encoder–Decoder Layer

The **Vision Transformer (ViT) [9]** forms the core of the proposed dehazing framework. Unlike CNN-based models that perform local convolutional filtering, the ViT employs global self-attention to model long-range dependencies and haze correlations across the image.

Patch Embedding

The seven-channel feature map $F(a,b,z)$ is divided into non-overlapping patches of size $P \times P$. Each patch is flattened and linearly projected into an embedding vector of dimension D:

$$z_p = W_E \cdot \text{Flatten}(F_p) + E_{\text{pos}} \tag{6}$$

where W_E represents the learnable embedding weights and E_{pos} denotes the positional encoding added to retain spatial information among the patches. This forms a sequence of N patch embeddings, where $N = (H \times W)/P^2$.

Transformer Encoder

Each patch embedding is passed through multiple stacked *Transformer encoder blocks*, consisting of *multi-head self-attention (MHSA)* and *feed-forward (FFN)* layers. The encoder learns to correlate haze patterns and illumination variations across the entire image domain. The transformation in the l^{th} encoder block is defined as:

$$Z_l = \mathrm{MHSA}(\mathrm{LN}(Z_{l-1})) + Z_{l-1} \tag{7}$$

$$Z'_l = \mathrm{FFN}(\mathrm{LN}(Z_l)) + Z_l \tag{8}$$

where LN denotes layer normalization, and Z_l and Z'_l represent the intermediate and final outputs of the l^{th} encoder layer, respectively.

Patch Reconstruction and Output Layer

After the encoder, the processed patch tokens are reshaped back into 2D spatial feature maps. A lightweight convolutional reconstruction head converts these features into the final three-channel RGB image:

$$\hat{J}(a,b) = f_{\text{recon}}(Z'_L) \tag{9}$$

where Z'_L denotes the final token set from the last Transformer block. Thus, the network learns a **direct mapping** between the multi-scale haze representation and the haze-free image:

$$\hat{J}(a,b) = f_{\text{ViT}}(F(a,b,z)) \tag{10}$$

The output $\hat{J}(a,b) \in \mathbb{R}^{H\times W\times 3}$ represents the restored dehazed image with accurate color, contrast, and luminance recovery.

1.4.3 Summary

The proposed **ViT-based dehazing network** leverages multi-scale physical priors and global attention mechanisms to achieve efficient and interpretable image restoration. Key highlights include:

- A seven-channel input derived from the Value (V) channel and multi-scale dark channels, encoding haze thickness across scales.
- A Vision Transformer backbone that performs global reasoning and learns haze removal end-to-end.
- A direct output of the dehazed RGB image without intermediate transmission computation.

This architecture combines the interpretability of physics-based priors with the modeling power of Transformers, yielding high-quality, color-consistent, and perceptually faithful dehazed images.

1.5 1.5 Ongoing Work and Future Plans

1.5.1 Ongoing Work

Currently, this research phase is focused on the implementation and experimental validation of the proposed dehazing model using a ViT-based approach. This involves work on the development of the complete coding pipeline, integrating the multi-scale feature extraction module into the backbone of the ViT, and then conducting its performance evaluation on different benchmark datasets.

The model will be trained and tested on two widely used datasets:

- **RESIDE-Outdoor (RESIDE-OTS):** for performance evaluation under natural outdoor hazy conditions.

- **RESIDE-Indoor:** to evaluate the generalization capability of the model in controlled synthetic haze environments.

Ongoing experiments are testing quantitative and qualitative performance using metrics like Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). The results will be compared against existing state-of-the-art dehazing networks, which include **DehazeFormer** [8] and **GridDehazeNet** [6], for validation of the effectiveness of the proposed architecture.

1.5.2 Future Plans

The next stage of this research will go beyond the basic framework of the Vision Transformer. Specifically, the future work will focus on more advanced transformer backbones such as:

- **Swin Transformer:** applied to enhance hierarchical feature aggregation and cross-scale attention.
- **Pyramid Vision Transformer (PVT):** to improve efficiency and incorporate both local and global contextual information.
- **Deformable Transformer architectures:** useful for modeling non-uniform haze distributions and capturing fine-grained structural details.

By incorporating these improvements, dehazing performance will be significantly enhanced in terms of spatial adaptability, reduced computational complexity, and stronger generalization across diverse haze conditions. Future extensions will be directed toward establishing a unified Transformer-based framework capable of handling multiple weather degradations such as rain, fog, and snow, inspired by **TransWeather** [7].

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Fig. 2 Plagiarism Report 1

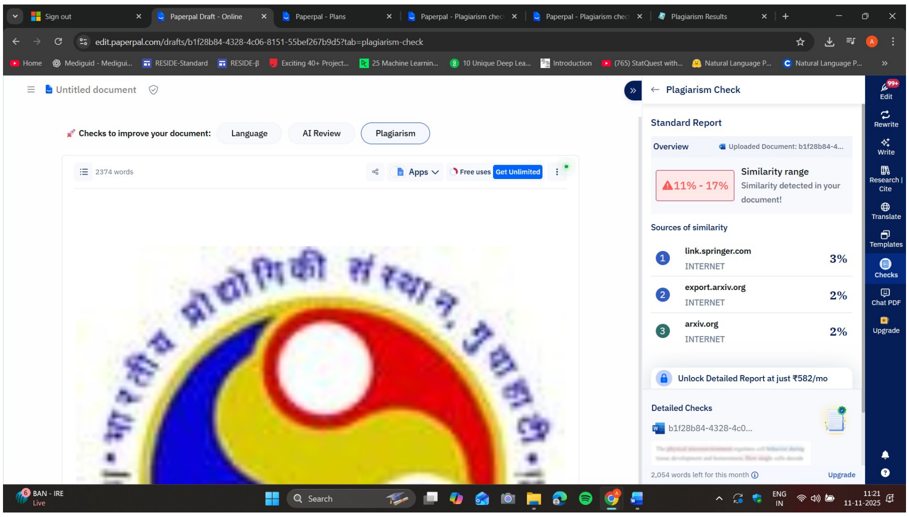

Fig. 3 Plagiarism Report 2

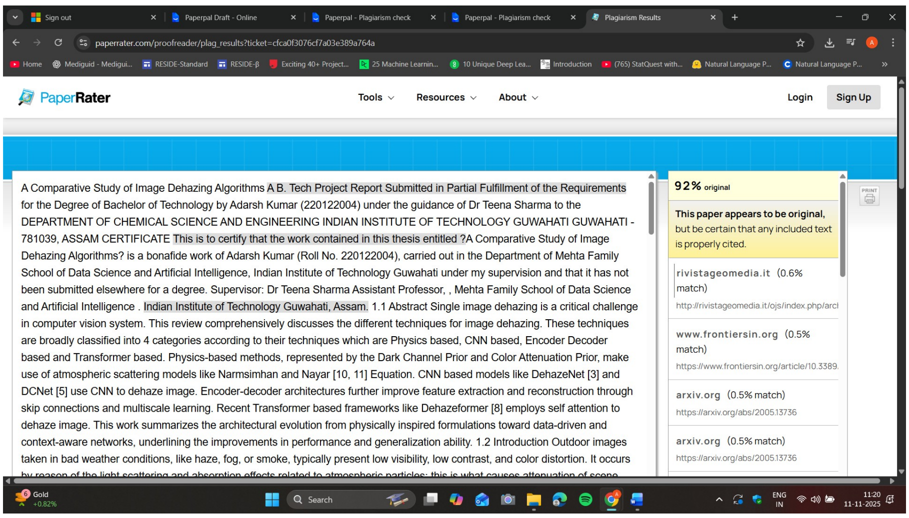

Fig. 4 Plagiarism Report 3